CONTINUATION-IN-PART (CIP)
SUPPLEMENTARY PATENT SPECIFICATION

CuraLive Platform

Third Submission to CIPC

Autonomous AI Self-Evolution Engine — Module 28

(Extending Modules 1–27 and Claims 1–33 of the Parent Provisional and CIP Submission 2)

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"System and Method for Autonomous AI Self-Evolution with Algorithmic Evidence Scoring, Cross-Event Correlation, and Autonomous Tool Promotion for Investor Communication Intelligence Platforms"

Cross-Reference to Related Applications:

Application ID	Filing Date	Status	Scope
1773575338868	12 March 2026	Filed (Parent Provisional)	Live Event Intelligence Platform (Modules 1–13 + 25 claims + 12 figures)
CIP Submission 2	16 March 2026	Filed	Autonomous Intelligence Modules 19–27 (8 claims 26–33 + 8 figures FIG 13–20)

This Filing (TBD)	18 March 2026	Pending	CIP Submission 3 — Module 28 (10 claims 34–43 + 7 figures FIG 21–27)

1. TITLE OF INVENTION

System and Method for Autonomous AI Self-Evolution with Algorithmic Evidence Scoring, Cross-Event Correlation, Evidence Decay Weighting, Autonomous Tool Promotion, Gap Detection Matrix, and Impact Estimation for Investor Communication Intelligence Platforms

2. ABSTRACT

This Continuation-in-Part (CIP) Submission 3 extends the parent provisional specification (Application ID 1773575338868, filed 12 March 2026) and the CIP Submission 2 (filed 16 March 2026) with one fundamentally new autonomous module: the Autonomous AI Self-Evolution Engine with Algorithmic Evidence Scoring (Module 28).

Unlike the operator-driven learning loops described in Module 9 (Self-Improving Communication Intelligence Models) and Module 13 (Autonomous Self-Evolving Platform Intelligence) of the parent specification, Module 28 operates with zero human input. The system autonomously observes the quality of its own AI-generated intelligence outputs, algorithmically identifies systematic weaknesses and capability gaps across multiple events and clients, correlates those gaps using cross-event pattern analysis, and autonomously proposes, scores, and promotes new AI tool capabilities through a five-stage lifecycle pipeline — all governed by six novel interconnected algorithms.

The six algorithms — Module Quality Scoring, Evidence Decay Function, Cross-Event Correlation Engine, Autonomous Promotion Pipeline, Gap Detection Matrix, and Impact Estimation Model — collectively form a closed-loop intelligence improvement system with no known prior art in any domain. Every event processed makes the system more aware of its own limitations, and that awareness automatically drives the creation, evidence-scoring, and promotion of new tool proposals.

3. FIELD OF THE INVENTION

The present invention relates to artificial intelligence systems for investor communication events and more particularly to methods for autonomous self-evolution of AI analysis platforms through algorithmic self-quality-assessment, evidence decay weighting, cross-event pattern correlation, and autonomous tool proposal promotion.

The invention further relates to closed-loop AI improvement systems that operate without human labelling, operator corrections, or supervised training signals — instead deriving all improvement signals from algorithmic evaluation of the system's own production outputs across multiple events, clients, and event types.

4. RELATIONSHIP TO PARENT AND PRIOR CIP SPECIFICATIONS

This CIP Submission 3 builds upon and extends the following prior filings:

Parent Provisional Specification (Application ID 1773575338868, filed 12 March 2026):

Disclosed 13 subsystems (Modules 1–13) with 25 independent claims and 12 figures covering real-time event monitoring, AI communication analysis, cross-platform capture via silent agents, OCC operator integration, anonymised benchmarking, autonomous compliance intervention, self-improving models, predictive intelligence, communication indexes, autonomous event intervention, and autonomous self-evolving platform intelligence.

CIP Submission 2 (filed 16 March 2026):

Extended the parent with 9 autonomous modules (Modules 19–27) adding 8 new independent claims (Claims 26–33) and 8 new figures (FIG 13–20). Modules 14–18 are operational expansion modules (Conference Dial-

Out, enhanced OCC, partner integrations, Intelligence Suite) that are part of the built platform but not separately claimed.

This CIP Submission 3 adds Module 28 (Autonomous AI Self-Evolution Engine with Algorithmic Evidence Scoring), contributing 10 new claims (Claims 34–43) and 7 new figures (FIG 21–27).

Key Differentiation from Parent Modules 9 and 13:

Module 9 (Self-Improving Communication Intelligence Models) of the parent specification described a system where operator corrections serve as training signals for model refinement — requiring human identification and submission of corrections. Module 13 (Autonomous Self-Evolving Platform Intelligence) described three-layer autonomy: real-time adaptation, inter-event learning, and platform evolution — but relied on operator feedback loops and supervised learning patterns.

Module 28 is fundamentally different: it operates with zero human input. The AI system itself determines where it is underperforming, what new capabilities would improve its outputs, and when sufficient evidence exists to promote a proposed capability from concept to approved implementation. No operator correction, labelling, or feedback is required at any stage.

5. BACKGROUND OF THE INVENTION (UPDATED)

The parent provisional specification and CIP Submission 2 collectively disclosed 27 subsystems with 33 independent claims and 20 figures, establishing a comprehensive platform for real-time investor communication intelligence. However, one critical capability gap remains unaddressed by prior art and by all prior filings:

No known system — in any domain — combines all of the following capabilities in a single closed-loop architecture:

1. Algorithmic self-quality-assessment of AI outputs across multiple quality dimensions (depth, breadth, specificity) without human labelling or evaluation.
2. Exponential time-decay weighting of self-improvement observations to prevent stale evidence accumulation and ensure only persistently observed gaps progress toward new tool creation.
3. Cross-event correlation for systematic gap detection that identifies platform-wide capability limitations distinguishable from event-specific anomalies, spanning multiple independent clients and event types.
4. Autonomous tool lifecycle management through a multi-stage promotion pipeline (emerging → proposed → approved → building → live) that operates without human gating at the first three stages.
5. A complete closed-loop cycle where events produce reports, reports are scored, scores become observations, observations cluster into proposals, proposals accumulate evidence, evidence drives promotion, and promoted tools improve future reports.

Prior Art Differentiation:

1. AutoML / Neural Architecture Search (Google, Meta): Optimises model hyperparameters and architectures. Module 28 observes the quality of AI outputs in a domain-specific context and proposes entirely new tools — not model tuning. The evidence decay, cross-event correlation, and autonomous promotion pipeline have no equivalent in AutoML.
2. MLflow / Weights & Biases / Neptune: ML experiment tracking platforms that log model metrics and compare runs. Module 28 does not track experiments — it autonomously identifies what new experiments should be created based on output quality analysis across live production data. No known tracking platform proposes new tools.
3. LangSmith / LangFuse (LLM observability): Trace prompt-response pairs and measure latency/cost. Module 28 goes beyond observability: it acts on quality observations by clustering them into tool proposals, scoring proposals with evidence decay, and promoting them through an autonomous lifecycle. Observability tools require humans to act on insights.
4. Chorus.ai / Gong (conversation intelligence): Generate fixed report types from conversation data. No known conversation intelligence platform evaluates the quality of its own reports, identifies systematic gaps across clients, and proposes new report types autonomously.
5. Reinforcement Learning from Human Feedback (RLHF): Requires human preference labels to improve model outputs. Module 28 requires zero human labels — it evaluates output quality algorithmically using the depth/breadth/specificity scoring method.

No known prior patent, product, or academic publication combines algorithmic self-quality-assessment, evidence decay weighting, cross-event pattern correlation, and autonomous tool proposal promotion in a closed-loop system for any AI intelligence platform.

6. SUMMARY OF THE INVENTION (UPDATED)

With this CIP Submission 3, the platform now comprises 28 interconnected subsystems: the original 13 from the parent specification (Modules 1–13), the operational expansion modules (14–18), the 9 autonomous modules from CIP Submission 2 (Modules 19–27), and the new Module 28 disclosed herein.

Module 28 — Autonomous AI Self-Evolution Engine with Algorithmic Evidence Scoring — introduces the final layer of platform autonomy: the ability for the AI system to evaluate, improve, and extend itself without any human input.

Module #	Subsystem Name	Source
1–13	Core Intelligence Platform (13 subsystems)	Parent (App ID 1773575338868)
14–18	Operational Expansion (Conference, OCC, Partners, Intelligence Suite)	Built platform (not separately claimed)
19	Autonomous Event Campaign Orchestrator	CIP Submission 2 (16 March 2026)
20	Self-Healing Regulatory Oracle	CIP Submission 2
21	Autonomous Investor Relationship Graph Weaver	CIP Submission 2
22	Self-Certifying Multi-Event Integrity Chain	CIP Submission 2
23	Autonomous AI Agent Swarm Commander	CIP Submission 2
24	Autonomous Valuation Impact Oracle	CIP Submission 2
25	Self-Auditing Investor Engagement ROI Engine	CIP Submission 2
26	Autonomous Dual-Purpose Compliance & Capital-Raising Advisor	CIP Submission 2
27	Self-Optimising Global Event Intelligence Network	CIP Submission 2
28	Autonomous AI Self-Evolution Engine with Algorithmic Evidence Scoring	This CIP Submission 3 (18 March 2026)

Module 28 comprises six interconnected algorithmic components, each independently novel and collectively forming a closed-loop intelligence improvement system:

Algorithm #	Name	Function
1	Module Quality Scoring	Evaluates every AI module output across depth, breadth, and specificity dimensions
2	Evidence Decay Function	Applies exponential time-decay weighting to observations (14-day half-life)
3	Cross-Event Correlation Engine	Detects systematic gaps spanning multiple clients and event types
4	Autonomous Promotion Pipeline	Five-stage lifecycle (emerging → proposed → approved → building → live)
5	Gap Detection Matrix	Ranks platform's weakest modules by importance, failure rate, quality, and breadth
6	Impact Estimation Model	Scores tool proposals across frequency, breadth, severity, and urgency

7. DETAILED DESCRIPTION OF MODULE 28 — AUTONOMOUS AI SELF-EVOLUTION ENGINE WITH ALGORITHMIC EVIDENCE SCORING

CIP SUBMISSION 3 — MODULE 28

Continuation-in-Part addition to parent provisional (App ID 1773575338868) and CIP Submission 2 (16 March 2026)

7.28 Autonomous AI Self-Evolution Engine with Algorithmic Evidence Scoring (Module 28)

Refer: *FIG 21–27*

Purpose: The AI autonomously observes the quality of its own intelligence outputs, algorithmically identifies systematic weaknesses and capability gaps, correlates those gaps across multiple events and clients, and autonomously proposes, scores, and promotes new AI tool capabilities without any human input or operator correction. This fundamentally extends Module 9 (Self-Improving Models) and Module 13 (Self-Evolving Platform Intelligence) from the parent specification by eliminating all human dependency from the improvement loop.

The system maintains two persistent data stores:

- Evolution Observations Table: Stores individual observations of AI output quality and capability gaps, each tagged with source event identifier, client name, event type, observation type, module name, confidence score, and suggested capability.
- Tool Proposals Table: Stores clustered tool proposals with lifecycle status, evidence count, average confidence, observation linkages, estimated impact label, and prompt templates for future tool generation.

7.28.1 Algorithm 1 — Module Quality Scoring

Refer: *FIG 25*

For each AI intelligence report generated by the platform (whether from a live event, uploaded archive, or pasted transcript), the Module Quality Scoring algorithm evaluates every intelligence module output across three independent quality dimensions:

Dimension 1 — Depth (weight: 0.40): Measures the volume and detail of the module output. For string outputs, depth is calculated as the ratio of character count to a baseline threshold of 600 characters. For array outputs, depth is calculated as the ratio of array length to a baseline of 5 items. For object outputs, depth is assessed by the presence of numeric values, substantive string values (exceeding 20 characters), and the count of populated fields.

Dimension 2 — Breadth (weight: 0.30): Measures the completeness of structured outputs. For array outputs, breadth is the ratio of items with two or more populated fields to total items. For object outputs, breadth is the ratio of non-null, non-empty fields to total fields.

Dimension 3 — Specificity (weight: 0.30): Measures whether the output references actual transcript content versus generic boilerplate language. The algorithm maintains a library of generic phrases commonly produced by language models when they lack sufficient context (including "the company," "management team," "key stakeholders," "various factors," "going forward," "not applicable," "insufficient data," and similar). Each occurrence of a generic phrase reduces the specificity score by a defined penalty factor:

```
specificity = max(0, 1 - (generic_phrase_count × penalty_factor))
```

The composite quality score for each module is:

```
composite = (depth × 0.40) + (breadth × 0.30) + (specificity × 0.30)
```

Each module has an assigned importance weight reflecting its value to investors and operators (e.g., board-ready summary: 0.95, compliance review: 0.95, investor signals: 0.90, ESG mentions: 0.50). A module is classified as "weak" when its composite score falls below:

```
weak_threshold = 0.30 × module_importance_weight
```

7.28.2 Algorithm 2 — Evidence Decay Function

Refer: *FIG 22*

The Evidence Decay Function applies an exponential time-decay weighting to all observations, ensuring that recent evidence of AI weakness carries significantly more weight than older observations. The decay function implements a configurable half-life (default: 14 days), meaning that an observation's influence on tool proposal scoring halves every 14 days:

```
decay_weight = 0.5 ^ (age_in_days / half_life_days)
```

The decayed evidence score for a collection of observations linked to a tool proposal is:

```
decayed_score = weighted_mean(confidence_i × decay_weight_i)
```

This mechanism serves two critical purposes:

1. It prevents stale observations from permanently inflating a tool proposal's evidence score. If a gap is resolved through improvements to existing modules, the evidence naturally decays and the proposal loses momentum.
2. It ensures that actively recurring gaps maintain strong evidence scores. If the same weakness continues to appear in new events, fresh observations replace decayed ones and sustain the proposal's promotion trajectory.

7.28.3 Algorithm 3 — Cross-Event Correlation Engine

Refer: *FIG 24*

The Cross-Event Correlation Engine detects patterns of AI weakness that span multiple independent events, clients, and event types. A gap that appears across different contexts represents a systematic platform limitation rather than an event-specific anomaly.

The correlation method operates as follows:

1. All non-weak-module observations are grouped by their suggested capability label or, if no label is assigned, by the first 60 characters of the observation text.
2. For each group with two or more observations, the engine calculates: client count (distinct client organisations), event type count, frequency, and average confidence.
3. A breadth score (0.0 to 1.0) is calculated for each group:

```
breadth = min(1.0, (clients-1)×0.30 + (event_types-1)×0.30 +
                   min(freq,10)/10×0.40)
```

4. Correlation results are ranked by the product of breadth score, average confidence, and frequency to produce a priority-ranked list of systematic platform gaps.

7.28.4 Algorithm 4 — Autonomous Promotion Pipeline

Refer: *FIG 23*

The Autonomous Promotion Pipeline manages the lifecycle of tool proposals through a five-stage progression without requiring human input at the first three stages:

Stage 1 — Emerging: A new tool proposal is created when the Accumulation Engine identifies a cluster of related observations that do not match any existing proposal.

Stage 2 — Proposed (auto-promoted): An emerging proposal is automatically promoted when it meets two threshold conditions simultaneously:

- Active evidence count (observations with decay weight above 0.25) reaches or exceeds 5
- Decayed evidence score reaches or exceeds 0.55

Stage 3 — Approved (auto-promoted): A proposed tool is automatically promoted when:

- Active evidence count reaches or exceeds 12
- Decayed evidence score reaches or exceeds 0.70

Stage 4 — Building: Reserved for tools that have been approved and are actively being implemented.

Stage 5 — Live: Reserved for tools that have been implemented and deployed to production.

A separate "rejected" status is available for proposals manually rejected by an operator. The demotion path is implicit through the evidence decay function: if a tool proposal's evidence stops being reinforced by new observations, the decayed score naturally drops below promotion thresholds, and the proposal stalls at its current stage.

7.28.5 Algorithm 5 — Gap Detection Matrix

Refer: *FIG 21*

The Gap Detection Matrix provides a systematic assessment of the platform's weakest intelligence modules by combining module importance, historical failure rate, current quality score, and the breadth of event types affected:

```
gap_score = importance_weight × failure_rate × (1 - quality) ×
                                breadth_factor
```

Where:

- module_importance_weight is the module's assigned weight (0.0 to 1.0)
- failure_rate is the ratio of times the module has been classified as weak to total evaluations
- current_quality is the module's most recent composite quality score
- breadth_factor is 1 + (distinct_event_types_affected × 0.10)

The matrix is sorted by gap score to produce a ranked identification of systematic intelligence blind spots across the platform's entire module grid.

7.28.6 Algorithm 6 — Impact Estimation Model

Refer: *FIG 26*

Every tool proposal receives a continuously updated impact score predicting the value of implementing the proposed tool. The impact model combines four equally weighted dimensions:

1. Frequency (weight: 0.25): The ratio of events producing observations supporting this proposal to total events analysed.

2. Breadth (weight: 0.25): breadth = min(1.0, (client_count × 0.40 + event_type_count × 0.30) / 3)
3. Severity (weight: 0.25): The average confidence score across all observations linked to the proposal.
4. Urgency (weight: 0.25): The ratio of recent observations (decay weight above 0.50) to total observations.

```
impact = (frequency × 0.25) + (breadth × 0.25) + (severity × 0.25) +
                                (urgency × 0.25)
```

Impact labels: Transformative (≥ 0.75) | High (≥ 0.55) | Medium (≥ 0.35) | Low (< 0.35)

7.28.7 The Accumulation Engine — Observation Clustering Method

The Accumulation Engine orchestrates observation clustering into tool proposals through the following process:

1. Retrieve all unclustered observations, ordered by recency, up to 200 per cycle.
2. Retrieve all existing tool proposals not in rejected or live status.
3. Run Cross-Event Correlation Engine (Algorithm 3) to identify highest-impact cross-event patterns.
4. Submit unclustered observations, existing proposals, and correlation patterns to a large language model to match observations to existing proposals or create new proposals.
5. For matches: link observation to existing proposal, merge into evidence list using set union, recalculate impact score using Algorithm 6.
6. For new proposals: create record with clustered observation IDs, initial evidence count, impact score, and prompt template for future tool generation.
7. Run Autonomous Promotion Pipeline (Algorithm 4) to evaluate all proposals for stage advancement.

7.28.8 The Meta-Observer — Automatic Invocation

The Meta-Observer is invoked automatically after every AI intelligence report generation, regardless of input method (live session, archive upload, or transcript paste):

1. Module Quality Scoring (Algorithm 1) scores all intelligence modules in the report across depth, breadth, and specificity dimensions.
2. Weak modules (those scoring below the importance-weighted threshold) are recorded as observations with full quality dimension data and source event metadata.
3. Gap Detection Matrix (Algorithm 5) is built using historical observation data and current quality scores to produce a ranked list of systematic blind spots.
4. A large language model is invoked with a fingerprint of the report (module scores, gaps, patterns) to identify missing capabilities, repeated patterns, and data gaps not detectable by algorithmic scoring alone.
5. All observations (both algorithmic and AI-identified) are persisted to the Evolution Observations Table with full metadata.
6. The Accumulation Engine is triggered asynchronously to process new observations, cluster them into proposals, and evaluate promotions.

This creates a fully autonomous feedback loop: every event processed makes the system more aware of its own limitations, and that awareness automatically drives the creation and promotion of new tool proposals. The closed-loop cycle is: events → reports → scoring → observations → proposals → evidence → promotion → improved reports.

7.28.9 Novel Aspects and Differentiation

1. Self-observation without human input: Unlike supervised learning systems that require labelled training data or operator corrections, this system evaluates its own output quality using the Module Quality Scoring algorithm. No human needs to identify weak output — the algorithm detects this automatically.
2. Evidence decay for self-correcting promotion: No known system applies exponential time-decay weighting to AI self-improvement observations. This prevents stale evidence accumulation and ensures only persistently observed gaps progress through the promotion pipeline.
3. Cross-event correlation for systematic gap detection: The correlation engine detects patterns across multiple independent events and clients — distinct from single-event analysis or A/B testing. A gap

appearing across different clients, event types, and time periods is a systematic platform limitation, not an anomaly.

4. Autonomous tool lifecycle management: The five-stage promotion pipeline operates without human gating at the first three stages. Proposals are created, promoted, and scored entirely by algorithms. Human involvement is only required for the building and deployment stages.

5. Closed-loop improvement: The combination of all six algorithms creates a complete autonomous cycle with no human in the loop at any evaluation, observation, clustering, scoring, or promotion stage. No known system implements this for AI intelligence platforms in any domain.

8. ALTERNATIVE IMPLEMENTATIONS

The systems and methods described in this supplementary specification may be implemented using a variety of computational architectures and analytical techniques.

The Module Quality Scoring algorithm may use any suitable quality assessment methodology including but not limited to: the three-dimension depth/breadth/specificity model described herein, information-theoretic quality measures, semantic similarity scoring against reference outputs, or hybrid approaches combining algorithmic and AI-based evaluation.

The Evidence Decay Function may use any suitable time-decay model including but not limited to: the exponential half-life model described herein, linear decay, polynomial decay, or sliding window approaches. The half-life parameter (default: 14 days) may be adjusted for different operational contexts.

The Cross-Event Correlation Engine may use any suitable pattern detection methodology including but not limited to: the breadth-score model described herein, graph-based community detection, hierarchical clustering, or embedding-based similarity analysis.

The Autonomous Promotion Pipeline may use any number of lifecycle stages and any suitable promotion threshold criteria. The five-stage model and specific thresholds described herein are provided for illustrative purposes.

The large language model components (Meta-Observer, Accumulation Engine) may utilise any suitable language model including but not limited to: GPT-4o, GPT-4o-mini, Claude, Gemini, Llama, or domain-specific fine-tuned models.

The specific examples, thresholds, and implementation details described in this specification are provided for illustrative purposes and should not be interpreted as limiting the scope of the invention to any particular technology, model, parameter value, or implementation approach.

9. NEW CLAIMS (CIP SUBMISSION 3)

The following claims are additional to and supplement Claims 1–25 of the parent provisional specification (Application ID 1773575338868) and Claims 26–33 of CIP Submission 2 (filed 16 March 2026).

Claim 34 (CIP): The system of claim 1 further comprising an autonomous AI self-evolution engine configured to: (a) score each module of every AI-generated intelligence report across three quality dimensions comprising depth, breadth, and specificity, each computed using module-type-specific algorithms with configurable weighting factors; (b) classify modules as weak when the weighted composite score falls below a threshold defined as a fraction of the module's assigned importance weight; (c) persist each weakness observation with the source event identifier, client name, event type, all three dimension scores, composite score, and classification reason; and (d) invoke a large language model to identify capability gaps not detectable by algorithmic scoring alone.

Claim 35 (CIP): The system of claim 34 further comprising an evidence decay function that applies an exponential time-decay weighting to all self-evolution observations using a configurable half-life parameter, wherein the decayed evidence score for a collection of observations is calculated as the weighted mean of individual observation confidence scores with weights defined by the formula: decay_weight = 0.5 ^ (observation_age_in_days / half_life_days), and wherein the decay mechanism causes tool proposals to stall or lose momentum when supporting observations are no longer being reinforced by new events, creating a self-correcting promotion pipeline.

Claim 36 (CIP): The system of claim 34 further comprising a cross-event correlation engine that: (a) groups non-module-quality observations by suggested capability label; (b) calculates for each group the number of distinct client organisations, distinct event types, observation frequency, and average confidence; (c) computes a breadth score combining client diversity, event type diversity, and observation frequency; and (d) ranks correlation results by the product of breadth score, average confidence, and frequency to identify platform-wide capability gaps distinguishable from event-specific anomalies.

Claim 37 (CIP): The system of claim 34 further comprising an autonomous promotion pipeline that advances tool proposals through a multi-stage lifecycle comprising at least the stages of emerging, proposed, and approved, wherein promotion from each stage to the next requires simultaneous satisfaction of a minimum active evidence count threshold and a minimum decayed evidence score threshold, and wherein the promotion evaluation executes automatically without human input after each accumulation cycle.

Claim 38 (CIP): The system of claim 37 wherein the promotion thresholds are: (a) from emerging to proposed: minimum 5 active evidence observations with decay weight above 0.25 and minimum decayed evidence score of 0.55; and (b) from proposed to approved: minimum 12 active evidence observations with decay weight above 0.25 and minimum decayed evidence score of 0.70.

Claim 39 (CIP): The system of claim 34 further comprising a gap detection matrix that calculates a gap score for each intelligence module using the formula: gap_score = module_importance_weight × historical_failure_rate × (1 - current_quality_score) × breadth_factor, where breadth_factor is calculated as 1 plus the product of distinct event types affected and a scaling coefficient, and wherein the matrix is sorted by gap score to produce a ranked identification of systematic intelligence blind spots across the platform's module grid.

Claim 40 (CIP): The system of claim 34 further comprising an impact estimation model that assigns each tool proposal a composite impact score calculated as the equally weighted mean of four dimensions: frequency of occurrence relative to total events analysed, breadth across distinct clients and event types, average severity of supporting observations, and urgency measured as the ratio of recent observations to total observations, wherein the composite score is classified into impact labels comprising transformative, high, medium, and low.

Claim 41 (CIP): A computer-implemented method for autonomous self-evolution of an artificial intelligence system for investor communication events, comprising the steps of: (a) generating an intelligence report containing multiple analysis modules from a communication event transcript; (b) algorithmically scoring each module across depth, breadth, and specificity quality dimensions; (c) identifying modules whose composite

quality score falls below an importance-weighted threshold; (d) recording each identified weakness as a structured observation with source event metadata and quality dimension data; (e) invoking a large language model to identify additional capability gaps not detectable by algorithmic scoring; (f) clustering observations into tool proposals using a combination of AI-powered semantic matching and cross-event correlation analysis; (g) applying exponential time-decay weighting to observation evidence; (h) autonomously promoting tool proposals through a multi-stage lifecycle based on evidence count and decayed score thresholds; and (i) repeating steps (a) through (h) for each subsequent intelligence report, creating a closed-loop self-improvement cycle.

Claim 42 (CIP): The method of claim 41 wherein the specificity dimension score is calculated by detecting the presence of generic language model output phrases in the module content and reducing the score proportionally to the count of detected generic phrases, thereby distinguishing between transcript-specific intelligence and boilerplate language model output.

Claim 43 (CIP): The method of claim 41 wherein the cross-event correlation step identifies capability gaps that appear across multiple distinct client organisations and multiple distinct event types, and assigns higher priority to gaps with greater client and event type diversity as measured by a breadth score combining client count, event type count, and observation frequency.

10. UPDATED DRAWINGS (NEW FIGURES FOR CIP SUBMISSION 3)

The following figures (FIG 21–27) supplement Figures 1–12 of the parent provisional specification and Figures 13–20 of CIP Submission 2.

Figure 21: AI Self-Evolution Engine: Complete System Architecture (Module 28)

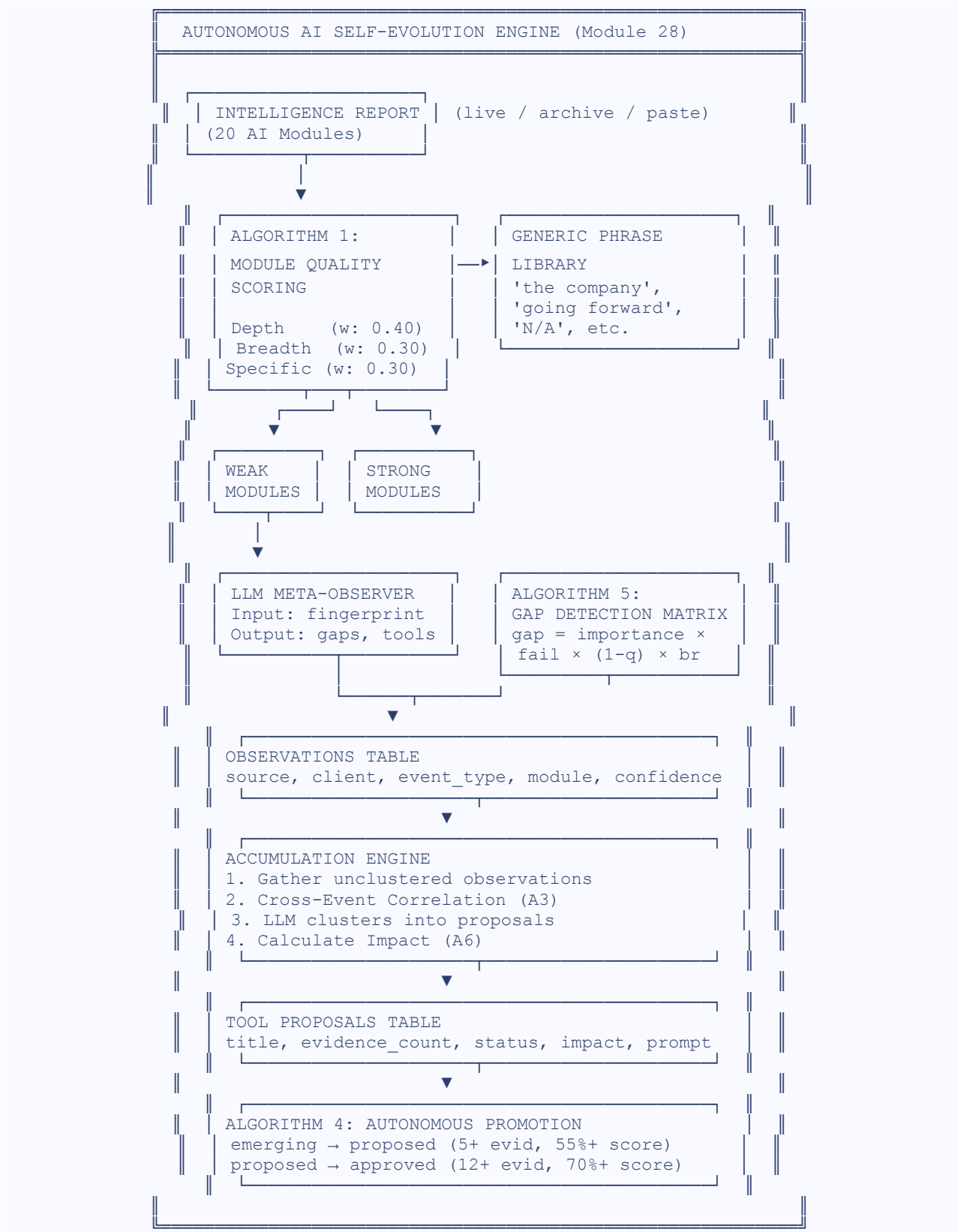


Figure 22: Evidence Decay Function: Time-Weighted Observation Scoring

```
                Observation Influence (decay_weight)
                                  |
                             1.00 |*
                                  |  *
                                  |     *
                            0.75 |        *
                                 |          *
         0.50 |                  *   <-- Half-life (14 days)
                            |                 *
    0.25 |                       * <-- Active evidence threshold
                     |                        *
    0.00 |_______________________ *____________________________
       0       7     14    21    28    35    42     49   56
                                   Age in Days

          Formula: decay_weight = 0.5 ^ (age_days / 14)

             Decayed Evidence Score (for proposal):
            sum(confidence_i × decay_i) / sum(decay_i)

                              Key:
         * Observations > 28 days contribute < 25% weight
         * Only observations with decay > 0.25 = 'active'
       * Proposals need sustained fresh evidence to promote
```

Figure 23: Autonomous Promotion Pipeline: Lifecycle State Machine

```
                    ┌──────────────┐
          ┌────►│  EMERGING    │
          │      │ < 5 obs      │
          │      │ score < 0.55 │
          │      └──────┬───────┘
          │             │
          │   [5+ evidence AND score ≥ 0.55]
          │             │
          │             ▼
          │     ┌──────────────┐────┐
          │     │  PROPOSED    │    │
          │     │ Operator can │    │
          │     │ reject       │    ▼
          │     └──────┬───────┘  ┌────────┐
          │            │          │REJECTED│
          │   [12+ evidence       └────┬───┘
          │     AND ≥ 0.70]            │
          │            │        (can restore)
          │            ▼              │
          │     ┌──────────────┐────┘
          │     │  APPROVED    │
          │     │ Ready for    │
          │     │ build        │
          │     └──────┬───────┘
          │            │
          │   [implementation started]
          │            ▼
          │     ┌──────────────┐
          │     │  BUILDING    │
          │     └──────┬───────┘
          │            │
          │   [deployed to production]
          │            ▼
          │     ┌──────────────┐
          │     │    LIVE      │
          │     └──────┬───────┘
          │            │
          │ NOTE: Evidence decay provides
          └── implicit demotion — proposals
              stall if evidence stops arriving
```

Figure 24: Cross-Event Correlation: Pattern Detection Across Clients

```
╔═══════════════════════════════════════════════════════════════════╗
  CROSS-EVENT CORRELATION ENGINE
╠═══════════════════════════════════════════════════════════════════╣
║                                                                   ╕
│  Client A       Client B      Client C      Client D             │
│  Earnings       AGM           Earnings      Board Mtg            │
   ╔═══╗          ╔═══╗         ╔═══╗         ╔═══╗
   ║   ╕          ║   ╕         ║   ╕         ║   ╕
   │   │          │   │         │   │         │   │
   │   │          │   │         │   │         │   │
   │   │          │   │         │   │         │   │
```

Figure 25: **Module Quality Scoring: Worked Examples**

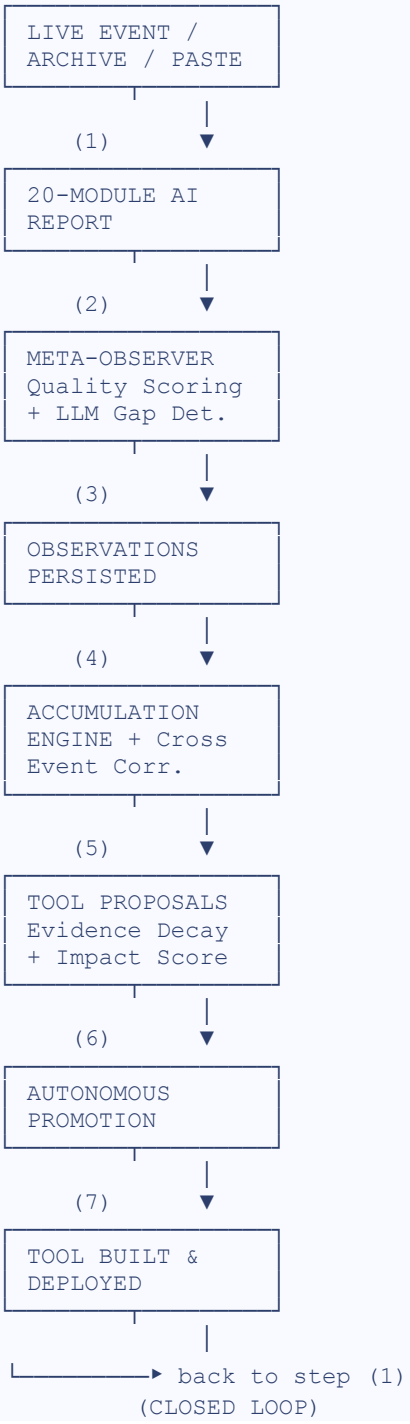
```
╔═══════════════════════════════════════════════════════════╗
│  EXAMPLE: 'Investor Signals' Module (weight: 0.90)        │
╠═══════════════════════════════════════════════════════════╣
│  Input: Array of 3 investor signal objects                │
│                                                           │
│   DEPTH     (w:0.40): len=3, depth=min(1, 3/5) = 0.60     │
│   BREADTH   (w:0.30): 2/3 items fully populated = 0.67    │
│   SPECIFIC  (w:0.30): 1 generic phrase, spec = 0.85       │
│                                                           │
│  COMPOSITE: (0.60×0.40)+(0.67×0.30)+(0.85×0.30) = 0.696   │
│   THRESHOLD: 0.30 × 0.90 = 0.27                           │
│   RESULT: 0.696 > 0.27 → NOT WEAK (passes)                │
╚═══════════════════════════════════════════════════════════╝


╔═══════════════════════════════════════════════════════════╗
│  EXAMPLE: 'ESG Mentions' Module (weight: 0.50)            │
╠═══════════════════════════════════════════════════════════╣
│  Input: Empty array []                                    │
│                                                           │
│   DEPTH: 0.00 | BREADTH: 0.00 | SPECIFICITY: 1.00         │
│   COMPOSITE: 0.30 | THRESHOLD: 0.15                       │
│   RESULT: 0.30 > 0.15 → NOT WEAK (low-weight passes)      │
╚═══════════════════════════════════════════════════════════╝
```

Figure 26: Impact Estimation Model: Four-Dimension Scoring

```
╔═══════════════════════════════════════════════════════════╗
║  PROPOSED TOOL: 'Peer Company Comparison Engine'          ╒
╒  Supporting observations: 8 | Events analysed: 15         ╒
╟═══════════════════════════════════════════════════════════╢
╒                                                           ╒
╒  FREQUENCY (w:0.25): 8/15 events = 0.533                  ╒
╒  BREADTH   (w:0.25): 4 clients, 3 types = 0.833           ╒
╒  SEVERITY  (w:0.25): avg confidence = 0.720               ╒
╒  URGENCY   (w:0.25): 6/8 recent = 0.750                   ╒
╒                                                           ╒
╒  COMPOSITE: (0.533+0.833+0.720+0.750)×0.25 = 0.709        ╒
╒                                                           ╒
╒  IMPACT LABEL: HIGH (≥ 0.55, < 0.75)                      ╒
╒                                                           ╒
╒  Labels:                                                  ╒
╒    Low (<0.35) | Medium (0.35-0.54) | High (0.55-0.74)    ╒
╒    | Transformative (≥ 0.75)                              ╒
╚═══════════════════════════════════════════════════════════╝
```

Figure 27: Closed-Loop Self-Evolution Cycle



11. INCORPORATION BY REFERENCE

The full specification, claims (1–33), drawings (FIG 1–FIG 20), definitions, detailed description, and all alternative implementations described in the parent provisional specification (Application ID 1773575338868, filed 12 March 2026) and CIP Submission 2 (filed 16 March 2026) are incorporated by reference in their entirety. The new matter described in this CIP Submission 3 supplements and extends the prior filings without modifying or replacing any of the parent claims or described embodiments.

12. COMPLETE CLAIMS SUMMARY

25 independent claims from the parent provisional + 8 CIP Submission 2 claims + 10 CIP Submission 3 claims = 43 total claims across all submissions.

Claim #	Type	Scope	Filing
1–15	System Claims	Core platform: event monitoring, AI analysis, cross-platform capture, OCC, benchmarking, compliance, self-improving models	Parent (App ID 1773575338868)
16–20	Method Claims	Cross-platform capture method, benchmarking method, collaborative management, self-improving models	Parent
21–25	Autonomous Claims	Self-evolving platform intelligence, pattern discovery, predictive briefings, benchmark evolution	Parent
26–33	CIP Claims	Autonomous Modules 19–27: Campaign Orchestrator, Regulatory Oracle, Integrity Chain, Swarm Commander, Valuation Oracle, ROI Engine, Compliance Advisor, Global Intelligence Network	CIP Submission 2 (16 March 2026)
34	CIP System	Autonomous AI Self-Evolution Engine — Module Quality Scoring (Module 28)	This CIP Submission 3
35	CIP System (dep. 34)	Evidence Decay Function with exponential half-life weighting	This CIP Submission 3
36	CIP System (dep. 34)	Cross-Event Correlation Engine for systematic gap detection	This CIP Submission 3
37	CIP System (dep. 34)	Autonomous Promotion Pipeline — multi-stage lifecycle	This CIP Submission 3
38	CIP System (dep. 37)	Promotion Thresholds — specific evidence count and score thresholds	This CIP Submission 3
39	CIP System (dep. 34)	Gap Detection Matrix — ranked module blind spots	This CIP Submission 3
40	CIP System (dep. 34)	Impact Estimation Model — four-dimension scoring	This CIP Submission 3
41	CIP Method	Computer-implemented method for autonomous AI self-evolution	This CIP Submission 3
42	CIP Method (dep. 41)	Specificity dimension — generic phrase detection	This CIP Submission 3
43	CIP Method (dep. 41)	Cross-event correlation method — breadth-scored gap detection	This CIP Submission 3

13. DECLARATION

I, David Cameron, the applicant and inventor, declare that the new matter described in this CIP Submission 3 (Module 28 — Autonomous AI Self-Evolution Engine with Algorithmic Evidence Scoring) was not disclosed in the parent provisional patent specification (Application ID 1773575338868, filed 12 March 2026) or CIP Submission 2 (filed 16 March 2026), that the new matter has been reduced to practice in a working implementation of the CuraLive platform, and that this submission is filed within the priority period of the parent application.

David Cameron

_____

David Cameron

Applicant and Inventor

Date: 18 March 2026

End of Continuation-in-Part Supplementary Specification — Third Submission